

Pittsburgh 311 Waste and Neighborhood Cleanliness

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Purpose and Main Finding

I built a 180-record corpus around waste and neighborhood cleanliness within our broader Pittsburgh 311 Municipal Service Resolution topic. I wanted the data to support a practical extraction task: identify the service focus, the type of information in each record, and any responsible department that the source explicitly names. The main result was that prompt recovery improved Phi's targeted service-focus field, but model capability made the larger difference in full-record agreement.

Corpus Construction

I used current City of Pittsburgh guidance together with historical WPRDC codebook rows and operational summaries. The final mix includes 106 text records and 74 table records, so 41.1% of the corpus is tabular. The city pages explain curbside collection, missed pickup, illegal dumping, hazardous waste, recycling, and property enforcement. The historical files add issue names, department labels, request counts, and closure-time summaries.

I kept source URLs, timestamps, license notes, and section context with every record. I also preserved legacy and ambiguous labels because they are part of the historical evidence. Removing them would make the corpus cleaner, but it would hide a real source of difficulty.

Model Evaluation

I finalized a random sample of 25 records before opening any Phi predictions. I reviewed each reference record under the same written rules and made the final labeling decisions using my own judgment. Codex assisted with the Python implementation and suggested a reproducible workflow. I directed the task scope, corpus criteria, extraction schema, evaluation design, failure analysis, and interpretation boundaries.

Run	Exact	Focus	Type	Department
Phi baseline	36%	72%	84%	64%
Phi recovery	40%	88%	80%	64%
GPT-5.6 Luna	88%	96%	92%	96%
TypeSafe JEV 1.13.0	72%	88%	88%	88%

Prompt Recovery

Phi sometimes treated any named organization as the responsible department. It also overused dumping for vague violations, commercial dumpsters, junk vehicles, and property-compliance records. I revised the prompt to require explicit evidence before assigning that label.

After the change, service-focus agreement improved from 72% to 88%. Two complete records were fixed, one previously correct record regressed, and exact agreement increased from 36% to 40%. Information-type agreement fell from 84% to 80%, while department agreement stayed at 64%. Since I designed and tested the change on the same 25 records, I treat this as a diagnostic result rather than evidence of generalization.

Corpus Structure

I named the six clusters Seasonal Recycling and Disposal Guidance, Historical Request Summaries, Graffiti and Legacy Codes, Missed Pickup Procedures, Hazardous Waste and Zero Waste Guidance, and Refuse Codebook Categories. Their representatives were coherent enough to support those names. Evaluation counts ranged from one to eight records per cluster, so I would use the clusters to select examples for review rather than rank their difficulty.

Hosted Comparison

GPT-5.6 Luna reached 88% exact agreement on the same 25 records with the baseline prompt and schema. Field agreement was 96% for service focus, 92% for information type, and 96% for responsible department. Luna still disagreed on three records. The result suggests that model capability mattered substantially for this task, but one small same-sample run is not a general benchmark. I also tested TypeSafe JEV 1.13.0 on the same records and target definitions. JEV reached 72% exact agreement and 88% agreement on each field. Its typed Choice and Noul interface used deterministic department candidates with complete candidate coverage, so I interpret the result as a complementary structured-decision experiment rather than an identical prompt comparison. Using prices published on September 21, 2026, the 25-record JEV run cost about \$0.00079 from API-reported input tokens, while the reconstructed Luna estimate was \$0.00282. JEV was about 72% cheaper and its observed mean request latency was 0.19 seconds versus 1.15 seconds for Luna; Luna's token count is estimated because the original run did not retain API usage fields.

Policy Implications

This workflow could support document indexing, taxonomy audits, and human-reviewed metadata extraction. Repeated disagreements are useful signals because they can reveal unclear guidance, taxonomy drift, or insufficient context. I would use the model to flag records for review rather than infer current routing from older department labels or send requests automatically.

Limitations

The corpus contains official guidance and historical tables, but it does not contain the original language residents used in 311 complaints. Current web pages and older codebooks also come from different periods, which creates taxonomy drift. Several historical rows have missing definitions, retired labels, conflicting identifiers, or department names that may no longer reflect current routing.

Administrative closure time records when a request changed status and does not prove that the physical issue was resolved. Twenty-five reference records provide a useful diagnostic sample, but they are too few to establish performance across every service type or neighborhood context.

What I Would Test Next

The next study should use a new held-out sample and actual resident complaint narratives, with representative coverage across neighborhoods and service types. I would also separate prompt development from final evaluation and review errors that could delay or misroute service. That design would test the intended intake use case more directly than the current document-extraction study.

Reflection

My main takeaway is that corpus construction shaped the evaluation as much as the model did. I had to decide which text units were meaningful, what historical ambiguity to retain, and which claims the sources could support.